



Integration of yolov5 with Siamese networks for enhanced microplastic detection through fluorescence microscopy images

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Abstract

Microplastics (MP) are persistent environmental pollutants that pose significant risks to ecosystems and human health. This study introduces a hybrid detection model that integrates YOLOv5 with a lightweight Siamese Neural Network (SNN) to enhance MP detection in fluorescence microscopy images. YOLOv5 identifies candidate regions, whereas the SNN performs refined classification using anchor-based learning, addressing challenges such as class imbalance and deformable shapes. Experimental results highlight the proposed model's superior performance, achieving precision (0.938), recall (0.930), F1score (0.927), and mean Average Precision (mAP50) 0.926, outperforming Mask Region-Convolutional Neural Network (MRCNN) (F1-score: 0.846, mAP50: 0.858) and Faster Region-Convolutional Neural Network (FRCNN) (F1-score: 0.568, mAP50: 0.865). Comparative analysis also showed the robustness of the model over YOLOv10 and YOLOv11, which achieved lower mAP50 values of 0.648 and 0.779, respectively. Leveraging the unique color-based properties of fluorescence microscopy, this approach provides a scalable and highly accurate solution for microplastic detection. Future work will focus on real-time applications and enhancing the generalizability across diverse datasets.

KEYWORDS: Siamese neural networks, YOLO, object detection, microplastics

INTRODUCTION

MP contamination poses a significant threat to human health and the environment owing to its persistence and widespread presence. Accounting for over 92.4% of ocean plastic debris [1], these particles disrupt ecosystems and affect marine life, with 90% of marine species consuming them, leading to bioaccumulation and toxin transfer [2]. Notably, MP is present in 94% of bottled water and 83% of tap water, highlighting its impact on human health [3]. Additionally, 32% of the 78 million



tons of plastic produced annually evade proper waste management, exacerbating its spread [4]–[5]. Detecting MP is crucial for understanding their prevalence, mitigating their effects, and protecting ecosystems and public health. Although thermal analysis (e.g., pyrolysis-GC-MS), spectroscopy (FTIR and Raman), and microscopy (optical and electron) are accurate, they have high cost and time commitments with notable limitations [6]. Processing and analyzing samples are labor-intensive procedures that require a great deal of physical labor and specialized knowledge. These methods require expensive equipment, are time-consuming, are not suitable for high-throughput analysis, cannot be used for routine monitoring, and have limited capacity to handle complicated or heterogeneous samples, which lowers their effectiveness in large-scale applications [7][8]. Artificial Intelligence (AI) has revolutionized this sector by addressing the limitations of traditional methods via automation of data processing, better detection accuracy, faster analyses, and efficient identification of MP in complex matrices [6]. MP identification algorithms based on deep learning include Recurrent Neural Networks (RNN) capable of handling sequential data of spectroscopic analyses and Convolutional Neural Networks (CNN) used to identify and classify MP in images [9]. Transfer learning further enhances the accuracy and efficiency of complex queries. In contrast, real-time detection and localization in image-based datasets are performed using advanced models, such as YOLO [10]. Using transfer learning with pre-trained architectures (such as ResNet and EfficientNet) significantly increases the accuracy and speed of complex tasks. Pytorch-based frameworks for FRCNN and MRCNN and ultralytics for the YOLOv5 model [10]. CNN evaluate pictures, and fluorescent imaging has emerged as a promising technique for enhancing the visibility of MP in environmental samples. Fluorescent imaging adds contrast by using UV light to emit fluorescence, whereas normal imaging uses visible light to record the appearance of MP based on reflection [11].

LITERATURE REVIEW

Several deep learning methods have been developed to identify MP in water. Grekov et al. [12] presented an MFFM FRCNN-based object detection model for identifying small residual film pieces on field surfaces. The model achieved superior performance with an 83.45% AP, 0.89 F1-score, and a detection speed of 248.36 ms, surpassing the original FRCNN, SSD, and YOLOv5. Its high accuracy and efficiency are pivotal for automating residual film removal processes. Xiao-Le Han et al., [13] applied MRCNN deep learning approach. They demonstrated high accuracy in the detection, classification, and segmentation of large marine MP particles, achieving high precision (93.3%), recall (95.4%), APbb (92.7%), F1 score (94.34%), and APm (82.6%) on a 250-image test dataset. This model was trained on 3000 MP images with a ResNet-101 backbone and processed images at 12.5 FPS.

Bianco et al. [14] developed an innovative method by combining a holographic setup with an SVM classifier (H-SVM), significantly surpassing the performance of traditional brightfield microscopy-based SVM (C-SVM), which had accuracies below 88%. The H-SVM method achieved an impressive accuracy of 99.31%, showing remarkable improvements in specificity (99.82% compared with 93%), sensitivity (97.22% compared with 91%), and overall accuracy (99.10% compared with 60.07%). Javier et al. [15] compared machine learning classifiers for counting and classifying MP, finding that the Random Forest classifier (RFC) outperformed others, including AdaBoost, K-NN, and C4.5, with an accuracy of 96.6%. These classifiers rely on 19 features, such as color-based metrics (RGB, HSV) and shape-based metrics (compactness and aspect ratio).

Hassan et al. [16] reported the state-of-the-art performance of MRCNN on the COCO dataset, where it outperformed earlier models in instance segmentation and bounding box detection, processing images at 5 FPS with high accuracy. R. Gandhi [17] highlighted the evolution of RCNN-based architectures, noting that RCNN was computationally slow (47 seconds/image) due to separate region proposals and classification. The FRCNN improved this to 0.3 seconds/image by integrating ROI pooling and further reduced it to 0.2 seconds/image with a Region Proposal Network (RPN). YOLO stands out for real-time applications, achieving 45–155 FPS, albeit with a slight trade-off in accuracy.

Minaee et al. [18] highlighted advancements in segmentation accuracy, showing that mean Intersection over Union (mIoU) on the PASCAL VOC dataset increased from 62.2% with FCN to 90.5% with EfficientNet+NAS-FPN. Similar improvements were noted in the Cityscapes dataset, where the latest models achieved a 23% relative gain over the FCN. Waheed et al. [19] compared MRCNN and FRCNN, noting that FRCNN achieved a mAP of 69.9% on the PASCAL VOC 2007 dataset, while MRCNN matched its precision while adding instance segmentation capabilities.

Ren et al. [20] demonstrated improvements in FRCNN by reducing the number of region proposals, achieving a mAP of 36.2% on PASCAL VOC and 42.0% on COCO datasets. K. He et al. [21] proposed MRCNN with a ResNet-50 backbone, achieving a mAP of 37.1% for instance segmentation on COCO and further improvements with Feature Pyramid Networks (FPN) and RoI Align for pixel-level alignment. YOLO-based models have also seen significant advancements. Owen Tamin et al. [22] developed a YOLOv5 model achieving 93.31% accuracy with RGB images and 92.96% when combining RGNIR data, showcasing the benefits of data fusion. Chengtao Cai [23] refined YOLOv5 architecture for marine garbage detection, achieving a mAP of 97.2% at an IoU threshold of 0.5. Mohamed Rabii Simou et al. [24] improved YOLOv5 for detecting plastic greenhouses from satellite images, with an optimal confidence value of 0.635 and an AP of 84.1%. Taoufik Saidani [25] introduced a YOLOv5 tiny model optimized for resource-constrained devices, achieving a mAP of 60.9% at IoU 0.5 and 44% across IoU thresholds from 0.5–0.95.

These advancements underscore the potential of deep learning models for MP detection. Models such as YOLOv5, MRCNN, and FRCNN have been widely adopted owing to their robustness and efficiency. Metrics such as mAP50, mean precision and mean recall are pivotal for evaluating model performance. Despite the deformable and inconsistent shapes of MPs, fluorescence microscopy aids detection by highlighting their distinct colors. These findings justify the selection of models, such as YOLOv5, YOLOv10, YOLOv11, MRCNN, and FRCNN, for detecting MPs in fluorescence microscopy images, leveraging their unique strengths in accuracy, speed, and segmentation.



Material and method:

Database info

For this study, an open-source dataset of fluorescence microscopy images of different variants of MP was used, namely polystyrene (PS), low-density polyethylene (LDPE), high-density polyethylene (HDPE), and PETE, with a dataset size of 662 images. The images were uniform, with a resolution of 640×640 . The total number of annotations from all images shows that PETE is heavily under-represented with only 137 annotations while HDPE is 4523, PS is 2179 and LDPE is 2108 which creates a significant imbalance in classes, to address this imbalance all the images with the annotations about PETE were dropped, thus the resultant dataset is of size 657 images and the class balance between HDPE, LDPE and PS still hold.

Reference: [<https://universe.roboflow.com/puttisan/microplastic-lstyl/dataset/1>]

Research flow map

This study aims to identify a model that performs best on the metrics mAP50, mean precision across all classes, mean recall across all classes, and overall F1-score. To achieve this, a comparative analysis was conducted of models YOLOv5, YOLOv10, YOLOv11, MRCNN, and FRCNN under uniform hardware conditions, while allowing framework-specific hyperparameters. In addition, analysis of the architecture of these models was done to understand the features that drive their performance. The proposed method addresses the identified gaps based on these insights.

General Architecture

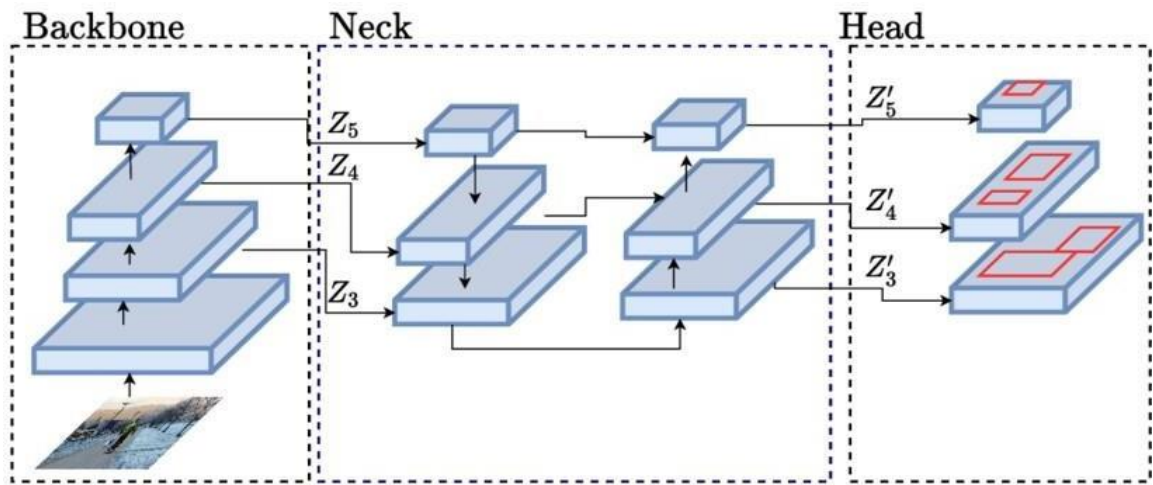


Figure 1: Generalized Architecture of a two-stage model

The YOLO and RCNN model families share a backbone, neck, and head architecture (Figure 1), differing primarily in their neck and head implementations. Backbones, usually deep CNNs, include ResNet50 for MRCNN and FRCNN and CSP Darknet53 for YOLO models. The FRCNN employs ROI and RPN pyramids to propose regions of interest, converting them into fixed-size feature maps. MRCNN extends the FRCNN by adding a mask branch for segmentation. YOLOv5 utilizes a Path Aggregation Network to enhance feature propagation and multiscale information, thereby strengthening its feature pyramid. Its detection head outputs class probabilities, bounding box coordinates, and object confidence scores.

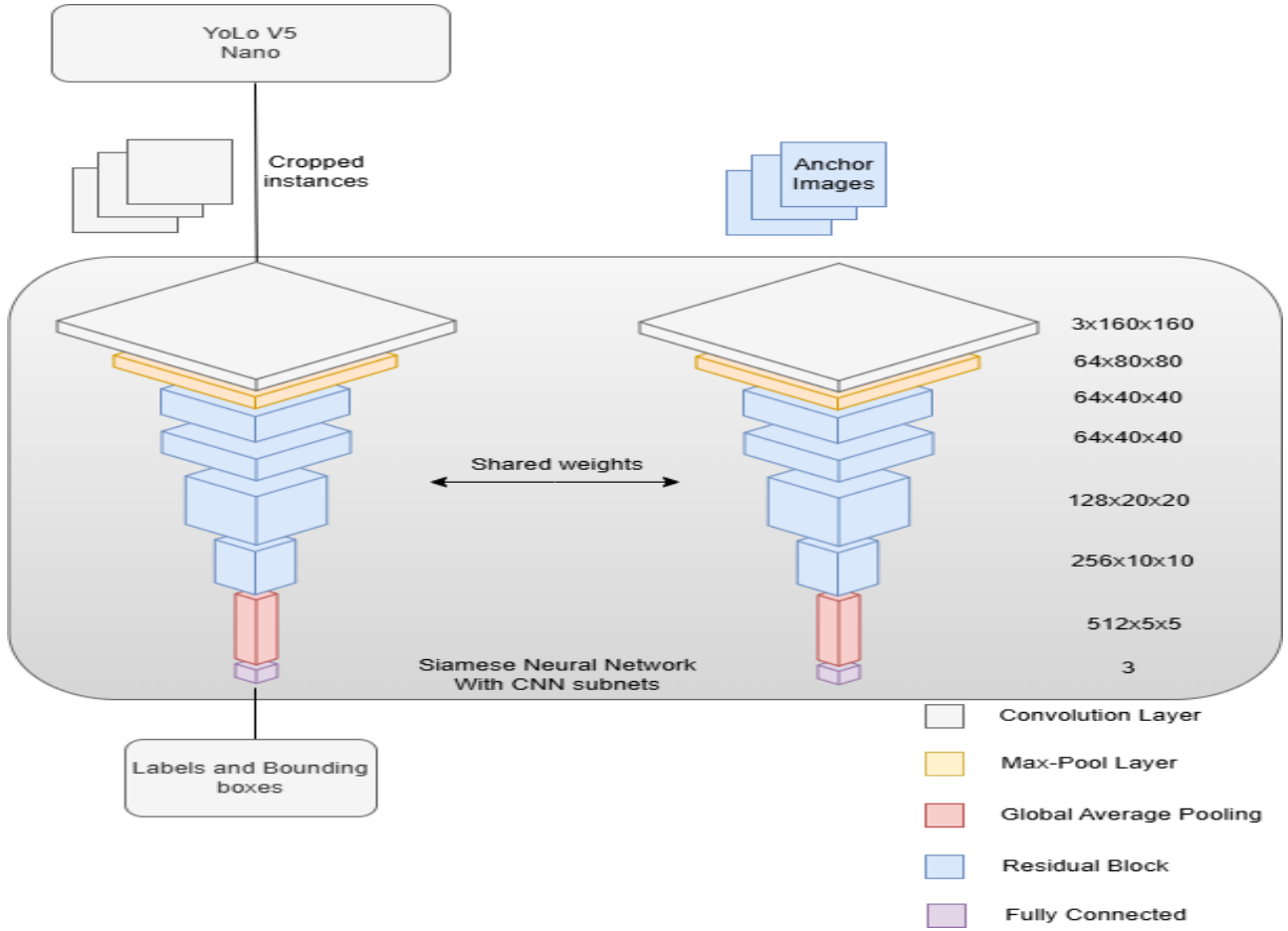


Figure 2: Anchor image of Siamese Neural Network with Resnet-18 subnets

The proposed method fine-tunes YOLOv5 to optimize box loss and detect it at a lower confidence threshold, potentially affecting the precision. To counter this, a lightweight SNN was introduced to classify cropped instances from YOLOv5's bounding boxes. The SNN uses three anchor images per class to label instances based on confidence scores and assigns the most common label with the highest confidence. Detections were discarded if the maximum confidence score for $n/2+1$ outputs was <0.5 or if no majority label was found, addressing false positives from the background.

Results and discussion

The models were trained using the Optuna framework for hyperparameter optimization while monitoring their validation losses. The weights with the least validation loss after the training were retained for the metrics shown in Table 1.

Model	Precision	Recall	F1 score	mAP50
Faster RCNN	0.467	0.727	0.568	0.865
Mask RCNN	0.765	0.947	0.846	0.858
YoloV5	0.849	0.831	0.840	0.875
YoloV10	0.733	0.573	0.643	0.648
YoloV11	0.692	0.889	0.778	0.779
SNN-YoLoV5	0.938	0.930	0.927	0.926

Table 1: Result analysis of different models

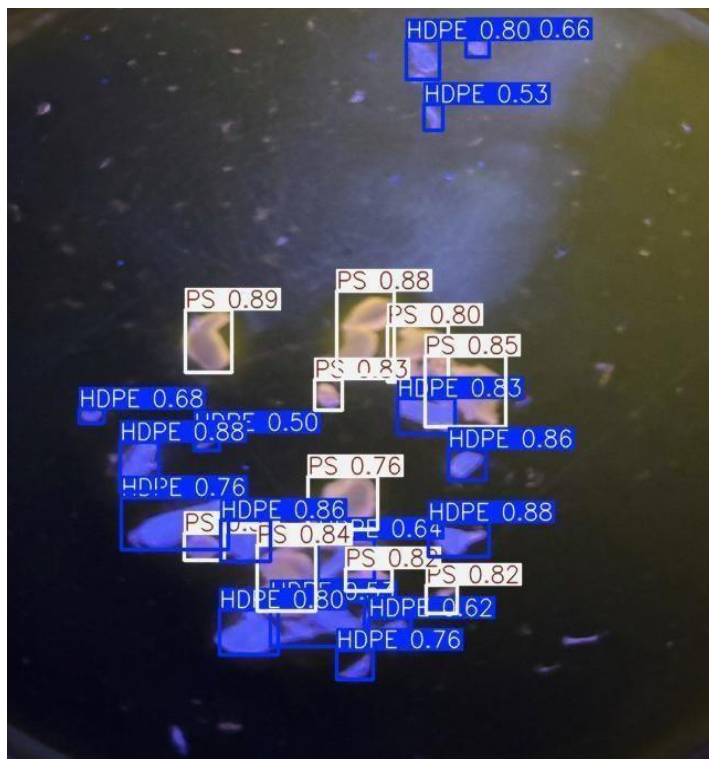


Figure3d



Figures 3a,3b,3c, and 3d show the outputs of the proposed model displaying multiple MP in the given samples.

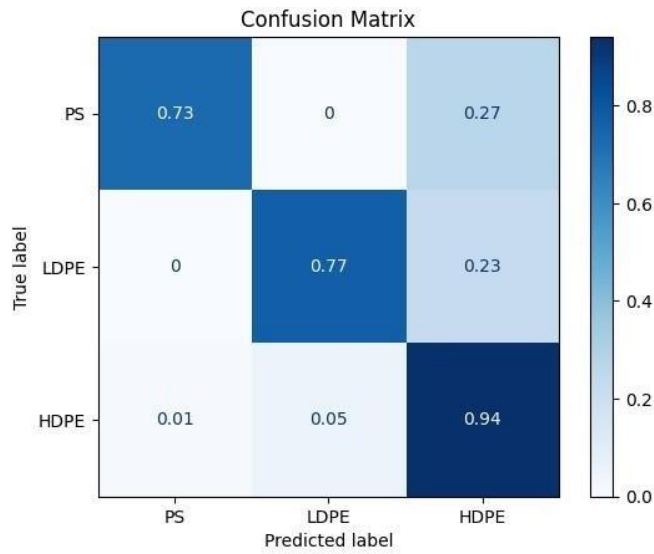


Figure 4: Mask RCNN Confusion Matrix

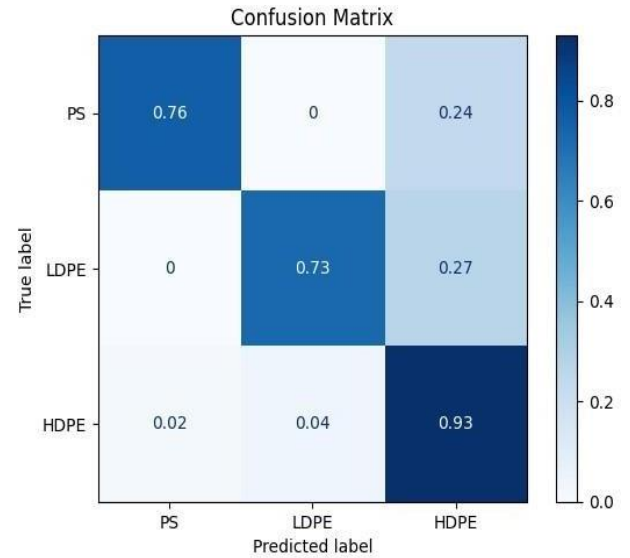


Figure 5: Faster RCNN Confusion Matrix

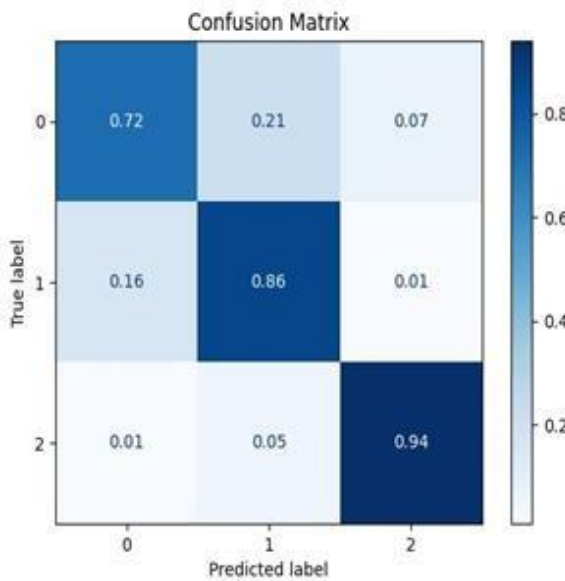


Figure 6: Yolov5 Confusion Matrix

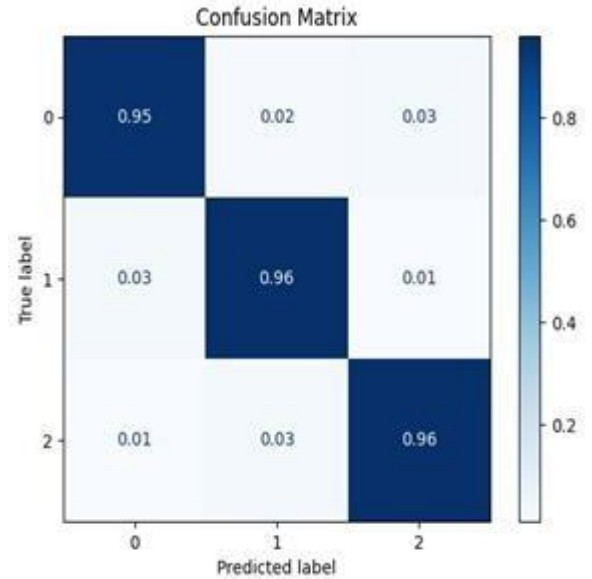


Figure 7: SNN-yolov5 Confusion Matrix



Confusion Matrix interpretation

When observing the confusion matrix and the results of Figure 4 MRCNN and Figure 5 FRCNN, it can be inferred that the two models mislabel PS as HDPE and LDPE as HDPE quite often, it can also be observed that the ratios on the matrices are the same but when looking at their recall it can be inferred that the number of true positives identified in MRCNN is far more compared to FRCNN. When considering Figure 6 YOLOv5, it can be observed that YOLOv5 mislabels PS as LDPE and mislabels LDPE as PS, a genuine confusion to behold, it is also to be considered that its precision and recall are both higher than those of the MRCNN and FRCNN. Based on the proposed method, it can be inferred that the confusion seen in just the vanilla YoLoV5 still holds, but faintly.

Model Results:

The proposed method Figure 7, shows an average gain in precision of 30% while showing a ~9% increase with the second-best model, an average gain in recall of 23% while showing a -1.7% decrease in recall, an average gain in F1 score of 27%, an 8% increase with the second-best model, an average gain in mAP50 of 21%, and a 5% increase with the second-best model.

Evaluation metrics

- mAP50: mAP at an IoU threshold of 0.5, which measures the precision of the object detection models across all classes. It was calculated by averaging the precision values at each class for detection with an IoU greater than 50%.
- True Positive (TP): A predicted bounding box is considered a True Positive if The IoU between the predicted box and the ground truth box was greater than or equal to a threshold. The expected box should correspond to the correct object (class-match).
- False Positive (FP): A predicted bounding box is considered a False Positive if The IoU between the predicted box and any ground truth box is less than the threshold (e.g., < 0.5). Or, the predicted box does not match any ground truth box, even if it has a high IoU with a ground truth box.
- False Negative (FN): A ground truth box does not match the predicted bounding box with IoU above the threshold (i.e., the model missed that ground truth).
- Mean Precision for All Classes: This metric calculates the average precision across all object classes, assessing the number of predicted positives that are correct.
- Mean Recall for All Classes: The average recall across all object classes measures how well the model detects all the relevant instances of each class.
- Overall F1 Score: The Harmonic means of precision and recall, providing a single measure that balances the two. It is especially useful when dealing with imbalanced datasets because it ensures that both false positives and false negatives are considered.

Role SNN-Yolov5 for MP detection

The proposed method considers three anchor images from the three classes PS, HDPE, and LDPE to allow the SNN subnetworks to learn similarities and differences. The YoLoV5 model detects the MPs in the image with a lower confidence threshold (0.4), and then the instances are extracted from the image using bounding boxes, which are paired with each instance, resulting in three pairs that are fed into the SNN to obtain the labels and confidence score for the given detection. If the maximum confidence score of the two outputs was < 0.5 , the detection was discarded. The label with the highest confidence score for a given output is selected, and the majority label is assigned if three distinct classes appear for the three outputs from the SNN, that is, no clear majority that detection is discarded. In this manner, it can detect almost all instances of MPs and can be precise. Because this is a pseudo-few-shot approach, the precision was higher.



Training process

The dataset is split into train and test sets in a split ratio of 70:30, YoLoV5's hyp.yaml is edited to increase the gain of the box loss function thus making it more sensitive to box loss changes and resulting in an optimized version in which box loss is minimal, the model was trained for 70 epochs, The default optimizers and loss functions were used which Ultralytics provides. A custom data loader class was implemented for the training of the SNN, where each sample in the data loader was a tuple of two images and a tuple of their metadata, which included the annotation boxes and corresponding labels. Upon initialization, the data loader crops the pictures into separate instances for training and resizes them, because performing these transformations when the `__getitem__` function is called creates a CPU bottleneck. The SNN was trained for 50 epochs using the Adam optimizer at a learning rate of 0.001, and a categorical cross-entropy loss function was used.

Qualitative and quantitative comparison, of different models

- **SNN-YoLoV5** is the best model for detecting microplastics in fluorescence microscopy images because of its high precision, recall, and mAP50. It is the most accurate method for detecting and segmenting MP with minimal false positives and nearly complete detection.
- **YoloV5** is also an excellent choice, offering high precision and a good balance between recall and precision, although it may miss a few microplastics compared with SNN-YoLoV5.
- **Mask R-CNN** performs well with high recall and good precision, making it a solid option, especially when segmentation accuracy is important.
- **Faster R-CNN** is less ideal owing to its lower precision, which could generate too many false positives, but it still has good recall and a decent mAP50.
- **YoloV10** and **YoloV11** showed lower recall and overall performance, making them less suitable for MP detection in fluorescence microscopy images, especially in tasks where detecting as many particles as possible is crucial.

Conclusions

This study successfully integrated YOLOv5 with a lightweight SNN to enhance the detection and classification of microplastics in fluorescence microscopy images. By leveraging YOLOv5's robust object detection capabilities and SNN's ability to discern fine-grained differences between classes, the proposed hybrid model achieves superior performance metrics compared to traditional and state-of-the-art models, such as FRCNN and MRCNN. The results demonstrate a significant improvement in precision (0.938), recall (0.930), and mAP50 (0.926), highlighting the method's efficiency in addressing challenges like class imbalance and deformable shapes of MP. The study also introduces a novel workflow that utilizes fluorescence microscopy's color distinction to aid in MP detection, a largely underexplored characteristic in prior research. Incorporating anchor-based learning into SNN enhances detection reliability while maintaining robustness against false positives. This approach offers a more accurate and scalable solution for MP detection and sets a foundation for extending the methodology to other applications that require high precision in object detection. Future work could focus on refining the SNN architecture for real-time applications and exploring additional data augmentation techniques to improve the generalizability across diverse datasets.

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